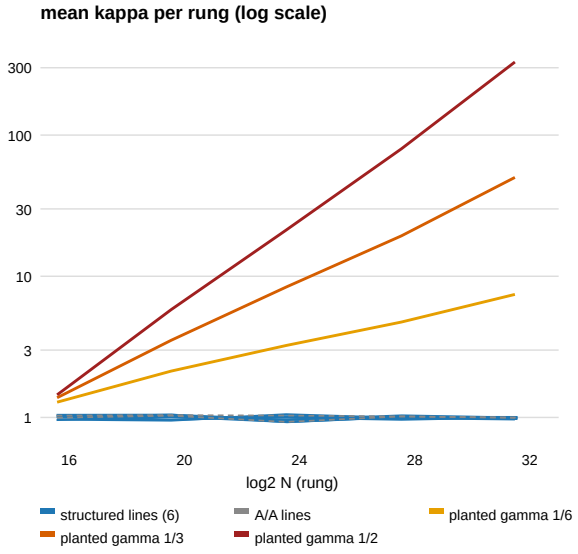
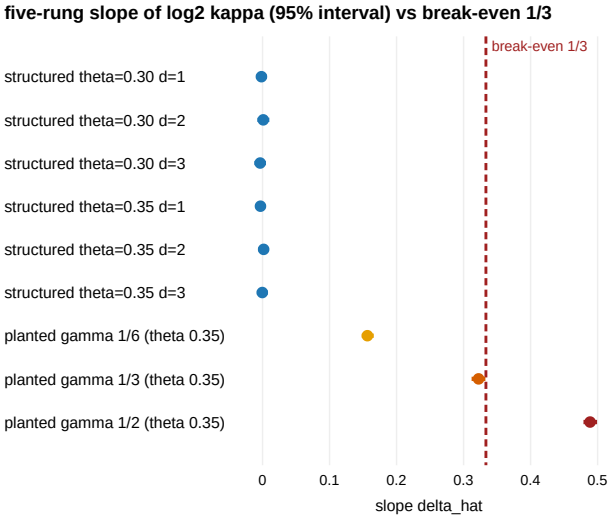
EXP-PFDR-2d36f9-R2: structured pair-sum coincidences do not grow with N (valid rerun)

Reading: `0-SUB-BE`. All six structured lines are valid, all blocking controls (C3 gamma-slope plants, C4 Z/N interval, C5 A/A lines, C6 z_cal calibration) pass, the selftest passes, no line is ALIVE, and no job failed (19,818 jobs; smallest rung mean coincidence count M = 3023 ≥ 2500). The relation excess κ of window and rank-d label-GAP symmetric bases stays at 1 across 16–32 bits: five-rung slopes of $\log_2\kappa$ on $\log_2 N$ are between −0.004 and +0.002, with the largest upper 95% limit 0.0101 against the break-even rate 1/3.

What this is. A rerun of `EXP-PFDR-2d36f9` (crypto-autoresearcher), which read O-INVALID on 2026-10-31 because its theta = 0.30 lines missed the powered floor (M = 2333 < 2500 at 16 bits) and its z_cal gate failed (sd 0.660 vs band [0.70, 1.30]). Two corrections were frozen before any R2 data (`PROTOCOL.md`, commit 534ff08faa): the sizing safety factor 1.5 instead of 1.1, and the calibration computed on the 10 independent null sets (the 30 cells share nulls across d; verified on the original raw data) with 95% bounds for n = 10. The kernel, curve code, cells, lines, statistics, controls, classes and readings are unchanged; fresh seeds.





Structured lines (κ per rung 16 / 20 / 24 / 28 / 32 bits; slope of $\log_2\kappa$ with 95% cluster-bootstrap interval)

line	class	slope [95%]	κ per rung
structured θ=0.30 d=1	SUB-BE	-0.0016 [-0.0099, 0.0066]	1.01 / 1.04 / 0.93 / 1.01 / 1.00
structured θ=0.30 d=2	SUB-BE	0.0011 [-0.0077, 0.0101]	0.96 / 1.01 / 0.93 / 1.01 / 0.98
structured θ=0.30 d=3	SUB-BE	-0.0036 [-0.0121, 0.0052]	1.04 / 1.04 / 0.96 / 1.03 / 1.00
structured θ=0.35 d=1	SUB-BE	-0.0031 [-0.0108, 0.0048]	1.02 / 1.00 / 1.00 / 0.96 / 1.00
structured θ=0.35 d=2	SUB-BE	0.0016 [-0.0064, 0.0097]	0.97 / 0.95 / 1.05 / 0.99 / 0.97
structured θ=0.35 d=3	SUB-BE	-0.0004 [-0.0085, 0.0079]	1.02 / 0.96 / 1.01 / 0.98 / 1.01

Controls

line	class	slope [95%]	κ per rung

plant θ=0.30 γ=0	SUB-BE	-0.0053 [-0.0125, 0.0018]	1.33 / 1.29 / 1.25 / 1.26 / 1.25
plant θ=0.30 γ=1/6	SUB-BE-GROWING	0.1581 [0.1524, 0.1639]	1.32 / 1.99 / 3.25 / 4.88 / 7.56
plant θ=0.30 γ=1/3	ALIVE	0.3308 [0.3251, 0.3365]	1.30 / 3.07 / 8.34 / 20.76 / 49.14
plant θ=0.30 γ=1/2	ALIVE	0.4991 [0.4936, 0.5045]	1.30 / 4.98 / 21.62 / 85.77 / 317.30
plant θ=0.35 γ=0	SUB-BE	-0.0042 [-0.0110, 0.0027]	1.33 / 1.33 / 1.27 / 1.31 / 1.26
plant θ=0.35 γ=1/6	SUB-BE-GROWING	0.1565 [0.1480, 0.1660]	1.28 / 2.13 / 3.23 / 4.74 / 7.44
plant θ=0.35 γ=1/3	SUB-BE-GROWING	0.3224 [0.3119, 0.3325]	1.38 / 3.52 / 8.38 / 19.35 / 50.17
plant θ=0.35 γ=1/2	ALIVE	0.4888 [0.4789, 0.4992]	1.45 / 5.79 / 21.25 / 80.02 / 329.50
Z/N θ=0.30 γ=—	ALIVE	0.6989 [0.6962, 0.7014]	1137.32 / 8025.74 / 55108.91 / 379293.77 / 2682371.03
Z/N θ=0.35 γ=—	ALIVE	0.6465 [0.6383, 0.6555]	673.68 / 3953.49 / 23596.76 / 142005.10 / 838712.51
A/A θ=0.30 d=1	SUB-BE	-0.0031 [-0.0115, 0.0058]	1.03 / 1.04 / 1.03 / 1.02 / 1.00
A/A θ=0.30 d=2	SUB-BE	-0.0031 [-0.0117, 0.0054]	1.03 / 1.04 / 1.03 / 1.02 / 1.00
A/A θ=0.30 d=3	SUB-BE	-0.0031 [-0.0114, 0.0053]	1.03 / 1.04 / 1.03 / 1.02 / 1.00
A/A θ=0.35 d=1	SUB-BE	-0.0005 [-0.0080, 0.0076]	1.00 / 1.03 / 0.93 / 1.02 / 0.99
A/A θ=0.35 d=2	SUB-BE	-0.0005 [-0.0079, 0.0073]	1.00 / 1.03 / 0.93 / 1.02 / 0.99
A/A θ=0.35 d=3	SUB-BE	-0.0005 [-0.0080, 0.0076]	1.00 / 1.03 / 0.93 / 1.02 / 0.99

The instrument sees growth: planted slopes γ = 1/6, 1/3, 1/2 are recovered as 0.158 / 0.322–0.331 / 0.489–0.499 (designs within the ±0.03 tolerance) and the Z/N interval reads 0.699 and 0.647 (expected 0.70 and 0.65); the γ = 0 plants and all six A/A lines read SUB-BE. **C6:** n = 10, mean 0.192 (≤ 0.62), sd 0.831 (in [0.548, 1.454]); reported only, the original 30-cell rule gives mean 0.192, sd 0.788, which would also pass the original bands [0.70, 1.30] and 0.40.

What it does and does not say

- Supports "no structural relation source of this family on F_p in 16–32 bits": the pair-sum coincidences of these structured bases track a random base at every rung, with no growth in N. Per the original specification the lead's disposition for these geometries is *exhausted at scale* with the obstruction max upper95(δ) = 0.0101 against 1/3. The disposition itself belongs to the sibling repository's process.
- Does not cover the FAM-1 or bbc53e-base ladders (the original specification defers them), nor other structured families, nor sizes above 32 bits. It prices no end-to-end index calculus and says nothing about rho.
- Run as a study in this repository (local, 4 workers, counts deterministic); it did not go through the sibling's coordinator approval and ledger, so its numbers are supplied for ingestion, not entered as a ledger decision. Raw

jobs (16 MB) are in the artifact store with a manifest.

- Consequence for the index-calculus gap: together with the residual-collector results (PR #3119: a generic collision search, floor about √(B+1) × rho), the only structural route in the lead atlas is now measured null with valid controls.